MATH 11 — ELEMENTARY STATISTICS

§6.2

Uniform Distributions

A constant probability density function

Today's Goal

After completing this section, you should be able to:

- Identify situations that can be modeled by a uniform distribution
- Find the probability density function for a uniform distribution
- Calculate probabilities for uniform distributions
- Find the mean and standard deviation of a uniform distribution

Learning Target 9

LT.9 Uniform Probability Distributions

I understand what a uniform probability distribution is. I can use a given uniform probability distribution to calculate the probability of an appropriate event.

Continuous Uniform Density Function

Definition 6.2.1

A **continuous uniform density function** is a probability density function whose output value remains the same for all values of the random variable. So

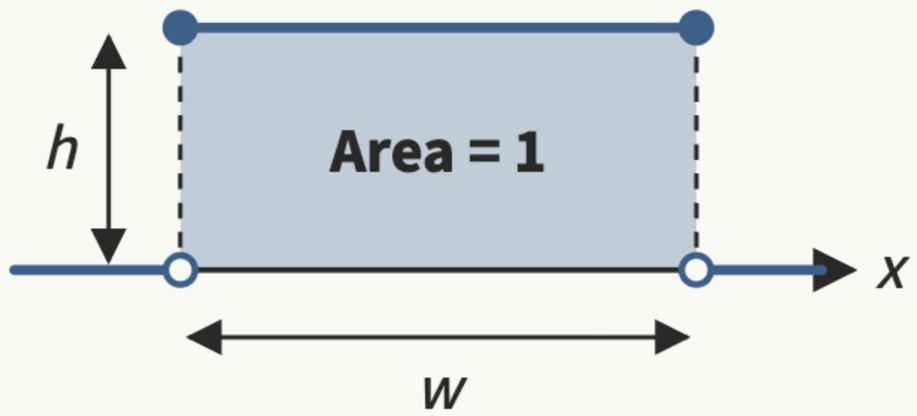
$$f(x) = h$$

for some constant value h.

The total area must equal 1. If the interval has length w, what is h?

The Height of the Density

The interval on which f is defined must have a finite length. If that interval has length w, then $h = \frac{1}{w}$ is the height of the density that makes the total area equal to 1.



Find the Density Function

A customer places an order with a food delivery app. The app says the order will arrive at any time between 20 and 45 minutes from now, with a uniform distribution. Let X represent the time (in minutes) until the order arrives. Find the continuous uniform density function $f(x)$ for X.

What are w and h?

The Density Function

The values of x range from 20 to 45, so the interval has length $w = 45 - 20 = 25$. Hence $h = \frac{1}{25}$, and the density is

$$f(x) = \begin{cases} 0 & \text{if } x < 20 \\ \frac{1}{25} & \text{if } 20 \leq x \leq 45 \\ 0 & \text{if } x > 45. \end{cases}$$

Uniform Distribution

Definition 6.2.3

A **(continuous) uniform probability distribution function** (uniform PDF) or "uniform distribution" for short, is a continuous probability distribution in which the probability density function is a uniform probability density function.

The notation $X \sim U(a, b)$ means that X is a random variable for a (continuous) uniform distribution where a is the lowest value of x and b is the highest value of x.

In Example 1, $X \sim U(20, 45)$.

Determining Uniform Probabilities

For a uniform distribution $X \sim U(a, b)$, the probability that X falls in an interval is the length of that interval divided by the total length $b - a$.

Theorem 6.2.4 Uniform Distribution Probability Formula

For a uniform distribution $X \sim U(a, b)$, the probability that X is between two values c and d (where $a \leq c < d \leq b$) is

$$P(c \leq X \leq d) = \frac{d - c}{b - a}.$$

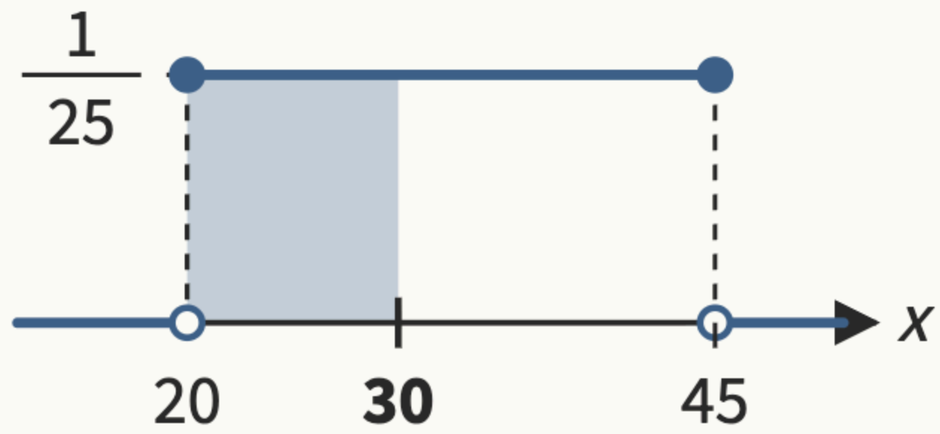
An Order Within 30 Minutes

For Example 1, what is the probability that the order will arrive within the next 30 minutes?

We found the density $f(x) = \frac{1}{25}$ for $20 \le x \le 45$.

We need $P(20 \le X \le 30)$, which is the area of the rectangle from $x = 20$ to $x = 30$:

$$P(20 \le X \le 30) = 10 \cdot \tfrac{1}{25} = \tfrac{2}{5}.$$



Now You Try: Uniform Probabilities

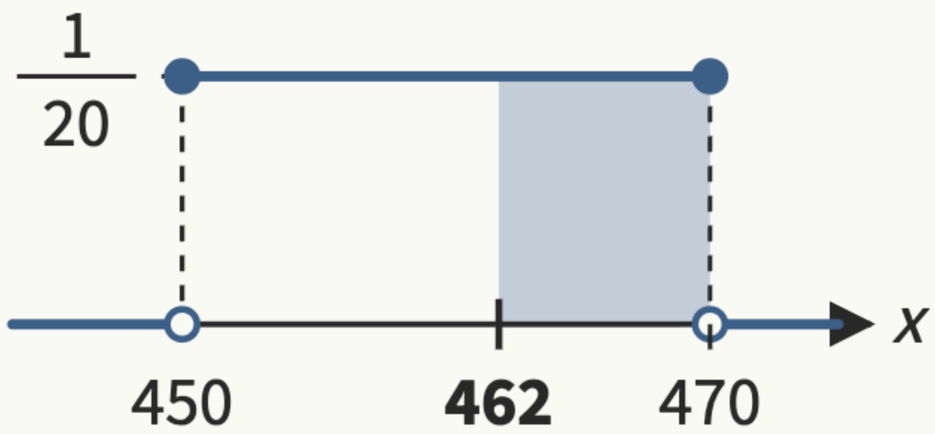
You Try 6.2.1. A bakery's ovens turn out sourdough loaves. Let X be the weight of a randomly chosen loaf, in grams. Assume that $X \sim U(450, 470)$.

Find the probability that the weight of a randomly chosen loaf is at least 462 grams. Round your answer to **four** decimal places.

You Try 6.2.1: Checking Your Answer

We have $X \sim U(450, 470)$, so the density is $f(x) = \frac{1}{20}$ for $450 \leq x \leq 470$.

We want $P(X \geq 462)$. Since X only takes values up to 470, this is the same as $P(462 \leq X \leq 470)$.



$$P(462 \leq X \leq 470) = 8 \cdot \frac{1}{20} = \frac{8}{20} = 0.4000$$

Parameters of a Uniform Distribution

For $X \sim U(a, b)$, where is the mean?

Theorem 6.2.6 Parameters of a Uniform Distribution

For a uniform distribution $X \sim U(a, b)$,

$$\text{Mean:} \quad \mu = \frac{a+b}{2}$$

$$\text{Standard Deviation:} \quad \sigma = \frac{b-a}{\sqrt{12}}.$$

Mean and Standard Deviation

For the food delivery scenario in Example 1, find the mean and standard deviation of the waiting time. Round to two decimal places if necessary.

We have $X \sim U(20, 45)$, so $a = 20$ and $b = 45$.

$$\mu = \frac{20 + 45}{2} = 32.5 \text{ minutes}$$

$$\sigma = \frac{45 - 20}{\sqrt{12}} = \frac{25}{\sqrt{12}} \approx 7.22 \text{ minutes}.$$

On average, a customer waiting for an order will wait 32.5 minutes for the order to arrive.

Now You Try: Parameters

You Try 6.2.2. A ride-share app sets the price of a ride from campus to the airport. Let X be the price of a randomly chosen ride, in dollars. Assume that $X \sim U(16, 30)$.

Find the mean and the standard deviation of X. Round each answer to **two** decimal places.

You Try 6.2.2: Checking Your Answer

We have $X \sim U(16, 30)$, so $a = 16$ and $b = 30$.

$$\mu = \frac{16 + 30}{2} = \frac{46}{2} = 23.00 \text{ dollars}$$

$$\sigma = \frac{30 - 16}{\sqrt{12}} = \frac{14}{\sqrt{12}} \approx 4.04 \text{ dollars}$$

On average, the price of a randomly chosen ride is 23 dollars.

Time to Finish a Quiz

Let X represent the time (in minutes) a student needs to finish a quiz. Then X is uniformly distributed with $X \sim U(8, 20)$.

(a) Find $P(X < 11)$.

$\frac{11-8}{20-8} = \frac{3}{12} = 0.25$

(b) Find $P(10 \leq X \leq 17)$.

$\frac{17-10}{20-8} = \frac{7}{12} \approx 0.5833$

(c) Find the mean and standard deviation.

$\mu = \frac{8+20}{2} = 14$ **minutes and** $\sigma = \frac{20-8}{\sqrt{12}} \approx 3.46$ **minutes**